

| 《数据库系统原理》实验报告（）                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |         |    |     |    |          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----|-----|----|----------|
| 题目：数据库实验                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |    |     |    |          |
| 学号                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2351136 | 姓名 | 李盛鹏 | 日期 | 2025.4.6 |
| <p>实验环境：</p> <p>硬件环境： windows\wsl, Docker Desktop 版本 28.0.1</p> <p>软件环境： OceanBase 社区版镜像:oceanbase/oceanbase-ce:lastest</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |    |     |    |          |
| <p>实验步骤及结果截图：</p> <p>1. 建表</p> <p>1.1 建立 Movie</p> <pre>Database changed obclient(root@sys)[mamba]&gt; create table Movie( -&gt;  Movie_no VARCHAR(10) NOT NULL, -&gt;  Movie_name VARCHAR(50) NOT NULL, -&gt;  Director VARCHAR(20), -&gt;  Rating FLOAT, -&gt;  End_date DATE, -&gt;  PRIMARY KEY (Movie_no) -&gt; ); Query OK, 0 rows affected (0.068 sec)</pre> <p>1.2 建立 Viewer</p> <pre>obclient(root@sys)[mamba]&gt; CREATE TABLE Viewer( -&gt;  Viewer_no VARCHAR(10) NOT NULL, -&gt;  Viewer_name VARCHAR(20) NOT NULL, -&gt;  Age INT, -&gt;  PRIMARY KEY (Viewer_no) -&gt; ); Query OK, 0 rows affected (0.026 sec)</pre> <p>1.3 建立 Watch</p> <pre>obclient(root@sys)[mamba]&gt; CREATE TABLE Watch( -&gt;  Viewer_no VARCHAR(10) NOT NULL, -&gt;  Movie_no VARCHAR(10) NOT NULL, -&gt;  Watch_date DATE, -&gt;  FOREIGN KEY(Viewer_no) REFERENCES Viewer(Viewer_no), -&gt;  FOREIGN KEY(Movie_no) REFERENCES Movie(Movie_no), -&gt;  PRIMARY KEY(Viewer_no, Movie_no) -&gt; ); Query OK, 0 rows affected (0.072 sec)</pre> <p>2.插入样例数据</p> <p>2.1 插入 Movie 数据</p> |         |    |     |    |          |

```
[172.17.0.2:2882] [2025-04-29 09:41:01.146455] [YB42AC110002-000633E79AE9D5F1-0-0]
obclient(root@sys)[mamba]> INSERT INTO Movie VALUES
-> ('M001', '星际穿越', '克里斯托弗·诺兰', 9.3, '2024-05-01'),
-> ('M002', '泰坦尼克号', '詹姆斯·卡梅隆', 9.1, '2024-04-10'),
-> ('M003', '盗梦空间', '克里斯托弗·诺兰', 8.8, '2024-04-20'),
-> ('M004', '科幻冒险之旅', '张三', 7.5, '2024-04-18'),
-> ('M005', '爱情故事', '李四', 7.0, '2024-04-25');
ERROR 1406 (22001): Data too long for column 'Director' at row 1
```

由于建表时设置的 **director** 有点小，所以没装下，因此做出修改

```
ERROR 1406 (22001): Data too long for column 'Director' at row 1
[172.17.0.2:2882] [2025-04-29 09:41:43.375113] [YB42AC110002-000633E79CD9D5EE-0-0]
obclient(root@sys)[mamba]> ALTER TABLE Movie MODIFY Director VARCHAR(50);
Query OK, 0 rows affected (0.059 sec)
```

## 2.2 插入 Viewer 数据

```
obclient(root@sys)[mamba]> insert into Viewer values
-> ('V001', '李明', 25), ('V002', '王红', 30), ('V003', '张磊', 22), ('V004', '赵颖', 28), ('V005', '孙阳', 35);
Query OK, 5 rows affected (0.006 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

## 2.3 插入 Watch 数据

```
obclient(root@sys)[mamba]> INSERT INTO Watch VALUES
-> ('V001', 'M001', '2024-03-15'),
-> ('V001', 'M002', '2024-03-20'),
-> ('V002', 'M002', '2024-03-25'),
-> ('V002', 'M003', '2024-04-01'),
-> ('V003', 'M001', '2024-04-05'),
-> ('V004', 'M002', '2024-04-12'),
-> ('V005', 'M003', '2024-04-14');
Query OK, 7 rows affected (0.009 sec)
Records: 7 Duplicates: 0 Warnings: 0
```

3. 查询电影名称中包含“科幻”的电影信息，输出所有信息（包括电影名称、电影编号、导演、评分、电影停映日期），并按照评分降序排列。

```
obclient(root@sys)[mamba]> SELECT * FROM Movie
-> WHERE Movie_name LIKE '%科幻%'
-> ORDER BY Rating DESC;
+-----+-----+-----+-----+-----+
| Movie_no | Movie_name      | Director | Rating | End_date |
+-----+-----+-----+-----+-----+
| M004     | 科幻冒险之旅    | 张三     | 7.5    | 2024-04-18 |
+-----+-----+-----+-----+-----+
1 row in set (0.000 sec)
```

4. 查询观看了电影名为“泰坦尼克号”的观众信息，输出该观众的编号、姓名和年龄，并按照观众编号升序排列。

```
obclient(root@sys)[mamba]> SELECT v.Viewer_no, v.Viewer_name, v.Age
-> FROM Viewer v
-> JOIN Watch w ON v.Viewer_no = w.Viewer_no
-> JOIN Movie m ON w.Movie_no = m.Movie_no
-> WHERE m.Movie_name = "泰坦尼克号"
-> ORDER BY v.Viewer_no ASC;
+-----+-----+-----+
| Viewer_no | Viewer_name | Age |
+-----+-----+-----+
| V001      | 李明       | 25  |
| V002      | 王红       | 30  |
| V004      | 赵颖       | 28  |
+-----+-----+-----+
3 rows in set (0.012 sec)
```

5.统计每个观众的观影信息，输出每个观众的编号、观看的电影名称和观看日期。

```
obclient(root@sys)[mamba]> SELECT v.Viewer_no, m.Movie_name, w.Watch_date
-> FROM Viewer v
-> JOIN Watch w ON v.Viewer_no = w.Viewer_no
-> JOIN Movie m ON w.Movie_no = m.Movie_no
-> ORDER BY v.Viewer_no;
+-----+-----+-----+
| Viewer_no | Movie_name   | Watch_date |
+-----+-----+-----+
| V001      | 星际穿越     | 2024-03-15 |
| V001      | 泰坦尼克号   | 2024-03-20 |
| V002      | 泰坦尼克号   | 2024-03-25 |
| V002      | 盗梦空间     | 2024-04-01 |
| V003      | 星际穿越     | 2024-04-05 |
| V004      | 泰坦尼克号   | 2024-04-12 |
| V005      | 盗梦空间     | 2024-04-14 |
+-----+-----+-----+
7 rows in set (0.003 sec)
```

6.查询所有已停映电影的信息，输出观众编号、姓名、电影名称和观看日期，并按观看日期降序排列。

```
obclient(root@sys)[mamba]> SELECT v.Viewer_no,m.Movie_name,w.Watch_date
-> FROM Viewer v
-> JOIN Watch w ON v.Viewer_no = w.Viewer_no
-> JOIN Movie m ON w.Movie_no = m.Movie_no
-> WHERE m.End_date < '2024-04-15'
-> ORDER BY w.Watch_date DESC;
+-----+-----+-----+
| Viewer_no | Movie_name   | Watch_date |
+-----+-----+-----+
| V004      | 泰坦尼克号   | 2024-04-12 |
| V002      | 泰坦尼克号   | 2024-03-25 |
| V001      | 泰坦尼克号   | 2024-03-20 |
+-----+-----+-----+
3 rows in set (0.003 sec)
```

7.查询观看了“星际穿越”但没有观看“盗梦空间”的观众信息，输出这些观众的编号，并按照编号升序排列。

```
obclient(root@sys)[mamba]> SELECT v.Viewer_no
-> FROM Viewer v
-> JOIN Watch w ON v.Viewer_no = w.Viewer_no
-> JOIN Movie m ON w.Movie_no = m.Movie_no AND m.Movie_name = '星际穿越'
-> WHERE NOT EXISTS(
->   SELECT 1 FROM Watch w1
->   JOIN Movie m1 ON w1.Movie_no = m1.Movie_no AND m1.Movie_name = '盗梦空间'
->   WHERE w1.Viewer_no = v.Viewer_no
-> )
-> ORDER BY v.Viewer_no ASC;
+-----+
| Viewer_no |
+-----+
| V001      |
| V003      |
+-----+
2 rows in set (0.005 sec)
```

8. 创建一个过程，使之能够实现如下功能：

9. 修改观影表，增加字段“重复观看状态”（字段名为“Repeat\_state”），字段含义为表示观众是否多次观看该电影；

```
obclient(root@sys)[mamba]> ALTER TABLE Watch ADD COLUMN Repeat_state BOOLEAN DEFAULT FALSE;
Query OK, 0 rows affected (0.030 sec)
```

10. 并根据表中已有数据为该字段赋值（所赋的值与表定义时的数据类型保持一致即可，比如可以定义多次观看某电影的“重复观看状态”为 True，只看过一次某电影的“重复观看状态”为 False），要求使用 if 语句进行条件判断。

```
obclient(root@sys)[mamba]> delimiter $$
obclient(root@sys)[mamba]>
obclient(root@sys)[mamba]> Create procedure update_repeat_state()
-> BEGIN
-> UPDATE Watch
-> SET Repeat_state = 'False';
->
-> UPDATE Watch w1
-> SET Repeat_state = 'True'
-> WHERE EXISTS(
->   SELECT 1 FROM Watch w2
->   WHERE w1.viewer_no = w2.viewer_no AND w1.movie_no = w2.movie_no AND w1.watch_date != w2.watch_date
-> );
-> END$$
Query OK, 0 rows affected (0.038 sec)
```

11. 在 8-10 题的基础上，查询没有重复观看过电影的观众信息，输出观众姓名和编号。

```
obclient(root@sys)[mamba]> SELECT DISTINCT v.Viewer_no, v.Viewer_name
-> FROM Viewer v
-> WHERE NOT EXISTS (
->   SELECT 1 FROM Watch w
->   WHERE w.Viewer_no = v.Viewer_no AND w.Repeat_state = TRUE
-> );
+-----+-----+
| Viewer_no | Viewer_name |
+-----+-----+
| V001      | 李明        |
| V002      | 王红        |
| V003      | 张磊        |
| V004      | 赵颖        |
| V005      | 孙阳        |
+-----+-----+
5 rows in set (0.015 sec)
```

出现的问题:

### 1. 插入表格失败

```
[172.17.0.2:2882] [2025-04-29 09:41:01.146455] [YB42AC110002-000633E79AE9D5F1-0-0]
obclient(root@sys)[mamba]> INSERT INTO Movie VALUES
-> ('M001', '星际穿越', '克里斯托弗·诺兰', 9.3, '2024-05-01'),
-> ('M002', '泰坦尼克号', '詹姆斯·卡梅隆', 9.1, '2024-04-10'),
-> ('M003', '盗梦空间', '克里斯托弗·诺兰', 8.8, '2024-04-20'),
-> ('M004', '科幻冒险之旅', '张三', 7.5, '2024-04-18'),
-> ('M005', '爱情故事', '李四', 7.0, '2024-04-25');
ERROR 1406 (22001): Data too long for column 'Director' at row 1
```

解决方案:

### 1. 更改表格中 director 的数据类型, 增加其大小

```
ERROR 1406 (22001): Data too long for column 'Director' at row 1
[172.17.0.2:2882] [2025-04-29 09:41:43.375113] [YB42AC110002-000633E79CD9D5EE-0-0]
obclient(root@sys)[mamba]> ALTER TABLE Movie MODIFY Director VARCHAR(50);
Query OK, 0 rows affected (0.059 sec)
```